Setting up a firewall using nftables

On linux system, there are tons of different tools to help you set up a firewall. In this lab we're going to do a brief overview of a tool called nftables which is a built in firewall tool on Kali Linux

Nftables setup

1. Launch your Kali Linux's terminal and make your root user.
2. Enable the nftables service so it'll always boot on startup by using the command **"systemctl enable nftables".** You'll also want to start the service by doing **"systemctl start nftables"**.

```
┌──(root㉿kali)-[/home/kali]
└─# systemctl enable nftables
Created symlink /etc/systemd/system/sysinit.target.wants/nftables.service → /
lib/systemd/system/nftables.service.

┌──(root㉿kali)-[/home/kali]
└─# systemctl start nftables
```

3. Check to see if the service is started and enabled by using the command **"systemctl status nftables"**

```
┌──(root㉿kali)-[/home/kali]
└─# systemctl status nftables
● nftables.service - nftables
     Loaded: loaded (/lib/systemd/system/nftables.service; enabled; preset: >
     Active: active (exited) since Wed 2024-07-24 12:12:55 EDT; 1min 37s ago
       Docs: man:nft(8)
             http://wiki.nftables.org
    Process: 9777 ExecStart=/usr/sbin/nft -f /etc/nftables.conf (code=exited>
   Main PID: 9777 (code=exited, status=0/SUCCESS)
        CPU: 13ms
```

4. Next we're going to create a table and associate it with IPv4 and IPv6 address using **"nft add table inet Firewall".** If you want to check to see if the tables are created. You can use the command **"nft list tables"**

```
┌──(root㉿kali)-[/home/kali]
└─# nft add table inet Firewall
```

5. Next we're going to add in policies to drop all traffic. You can do so using these two commands:
   nft 'add chain inet Firewall input { type filter hook input priority 0 ; policy drop ; }'

   nft 'add chain inet Firewall output { type filter hook output priority 0 ; policy drop ; }'

```
┌──(root㉿kali)-[/home/kali]
└─# nft 'add chain inet Firewall input { type filter hook input priority 0 ;
policy drop ; }'

┌──(root㉿kali)-[/home/kali]
└─# nft 'add chain inet Firewall output { type filter hook output priority 0
; policy drop ; }'
```

   Question for those doing the lab: Why do we need two? Also how can we check that this is implemented properly if we don't know the commands to check?

6. Then we're going to make our firewall connections stateful. Which we can do so using these two commands:
   nft add rule inet Firewall input ct state established,related accept
   nft add rule inet Firewall output ct state established,related accept

```
┌──(root㉿kali)-[/home/kali]
└─# nft add rule inet Firewall input ct state established,related accept

┌──(root㉿kali)-[/home/kali]
└─# nft add rule inet Firewall output ct state established,related accept
```

Allowing traffic:

1. Now that you've set up the firewall to drop all traffic and except traffic that is part of an ongoing communication that was previously allowed. We can now start explicating allowing protocols and ports. We'll first allow DNS using the commands:

nft add rule inet Firewall output udp dport 53 ct state new,established accept

nft add rule inet Firewall input udp sport 53 ct state established,related accept

```
┌──(root㉿kali)-[/home/kali]
└─# nft add rule inet Firewall output udp dport 53 ct state new,established a
ccept

┌──(root㉿kali)-[/home/kali]
└─# nft add rule inet Firewall output udp dport 53 ct state new,established a
ccept
```

2. Next we'll allow for https and http traffic. To do so we're going to use similar commands as the one we used for DNS:
nft add rule inet Firewall output tcp dport 443 ct state new,established,related accept

nft add rule inet Firewall input tcp sport 443 ct state established accept

nft add rule inet Firewall output tcp dport 80 ct state new,established accept

nft add rule inet Firewall input tcp sport 80 ct state established,related accept

```
┌──(root㉿kali)-[/home/kali]
└─# nft add rule inet Firewall output tcp dport 443 ct state new,established,
related accept

┌──(root㉿kali)-[/home/kali]
└─# nft add rule inet Firewall input tcp sport 443 ct state established accep
t

┌──(root㉿kali)-[/home/kali]
└─# nft add rule inet Firewall output tcp dport 80 ct state new,established a
ccept

┌──(root㉿kali)-[/home/kali]
└─# nft add rule inet Firewall input tcp sport 80 ct state established,relate
d accept
```

3. If you want to check if you created everything properly. You can open up your browser and see that you're now able to access webpages but if you tried to ssh

into another machine you won't be able to. Do you know why?

Also another way to check is to type **"nft list rulset"**

```
┌──(root㉿kali)-[/home/kali]
└─# nft list ruleset
table inet Firewall {
        chain input {
                type filter hook input priority filter; policy drop;
                ct state established,related accept
                ct state established,related accept
                tcp sport 443 ct state established accept
                tcp sport 80 ct state established,related accept
        }

        chain output {
                type filter hook output priority filter; policy drop;
                ct state established,related accept
                udp dport 53 ct state established,new accept
                udp dport 53 ct state established,new accept
                tcp dport 443 ct state established,related,new accept
                tcp dport 80 ct state established,new accept
        }
}
```